

A spectro-polarimetric view into accretion plasma physics

EXACTUS Seminar Day

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24.05.2024, University of Turku

Outline

1. Introduction to compact objects and X-ray binaries
2. Introduction to polarization and polarimetry
3. Applications and results
4. Conclusions

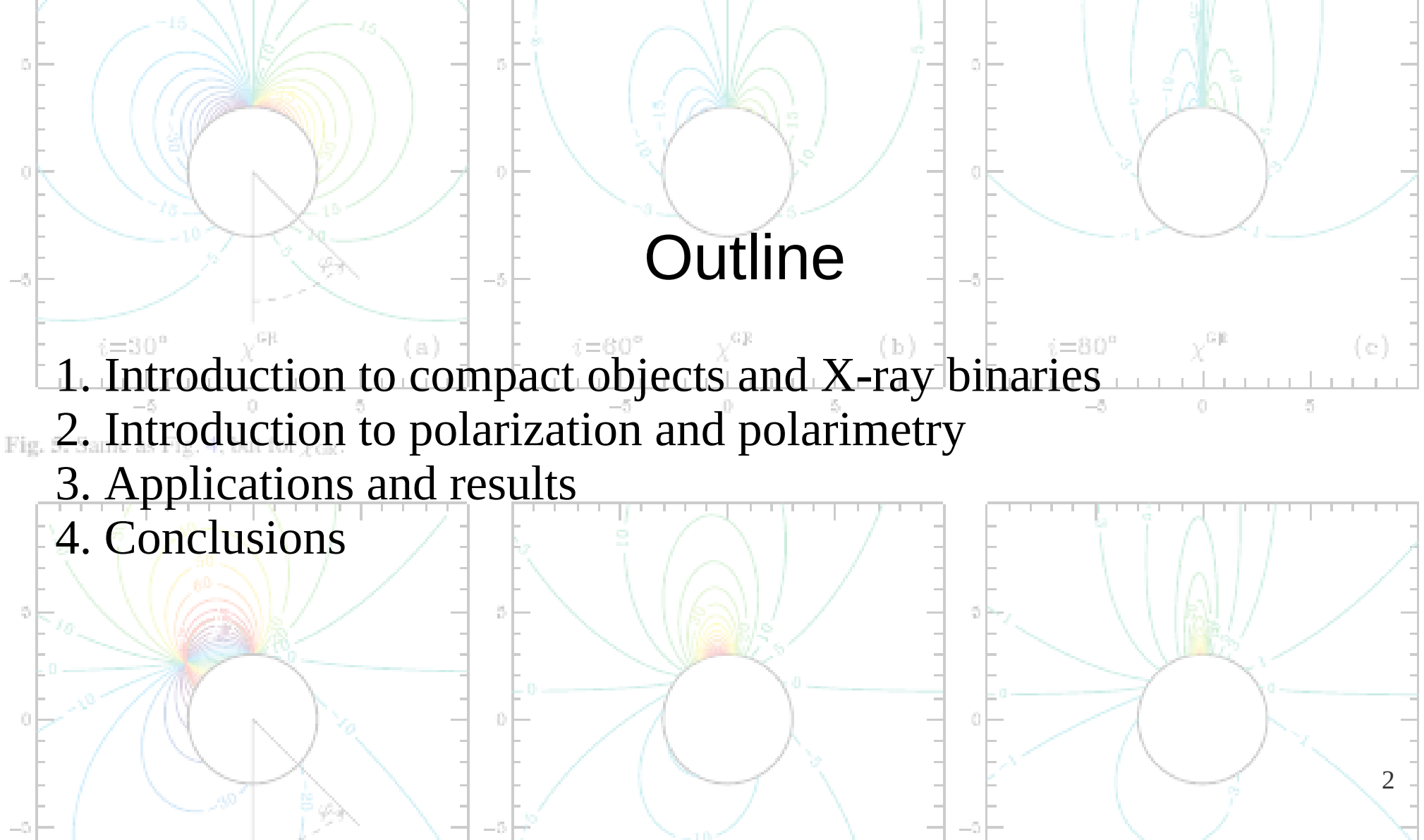
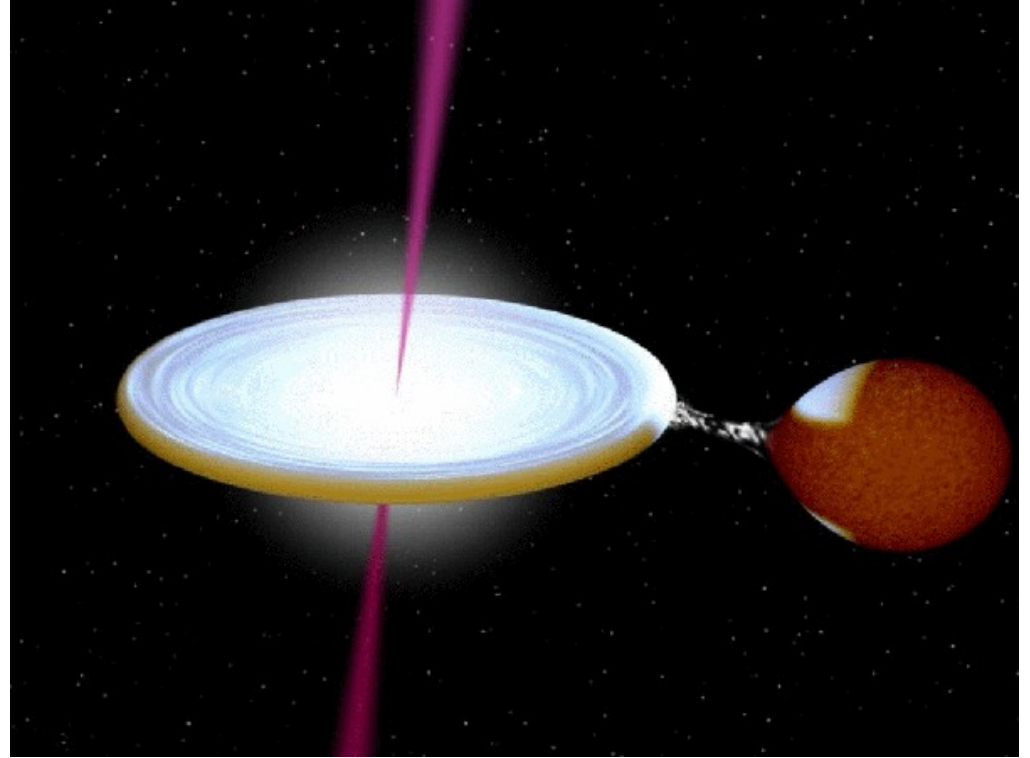


Fig. 5 (continued from Fig. 4) Plot of χ^{GR}

Compact objects in X-ray binaries

- Most stars out there are in gravitationally binary systems.
- One star most certainly dies before another; star death results in a compact object
- Such systems still interact in fascinating ways: forms relativistic flows, disks, jets, reflections etc.

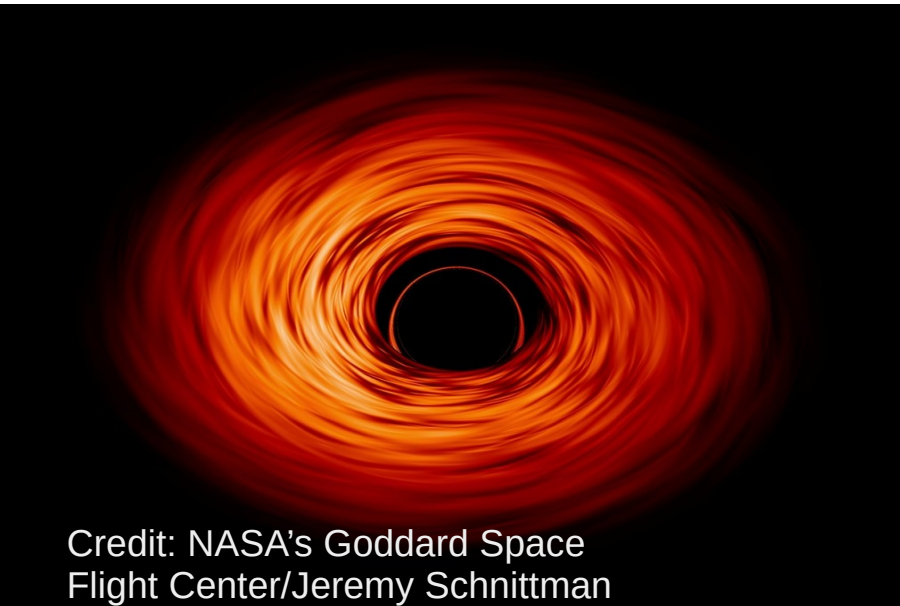


[An illustration of X-ray binary system GRS 1915. Credit: Rob Hynes.]

Types of compact objects

Black holes

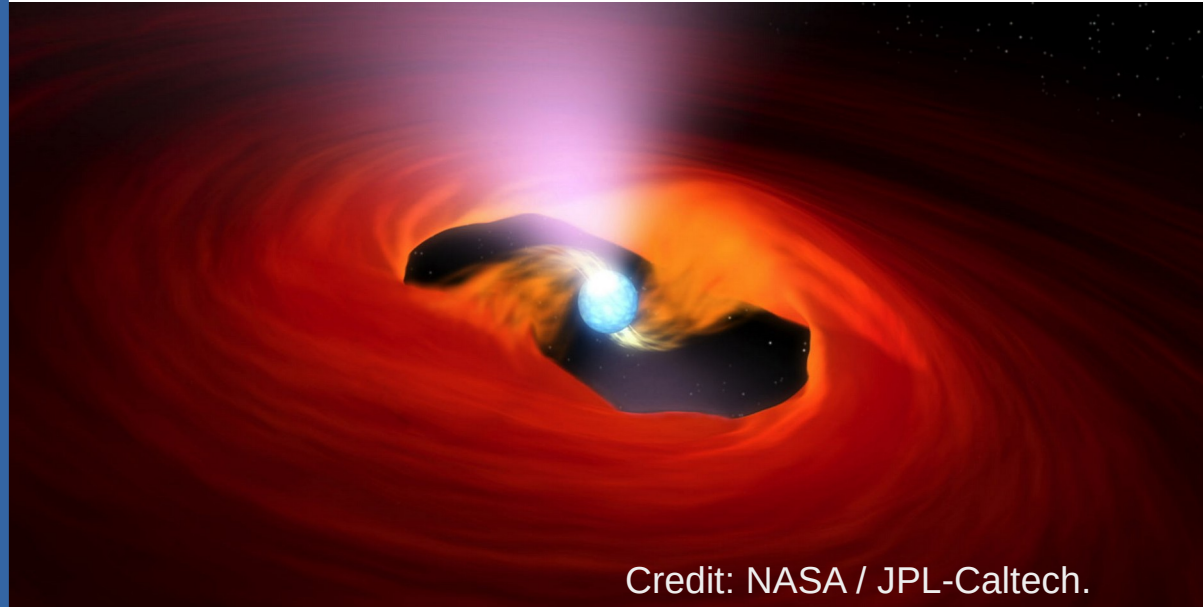
- Simple structure and M/R relation
- No hard surface, event horizon
- Angular momentum is infamously uncertain



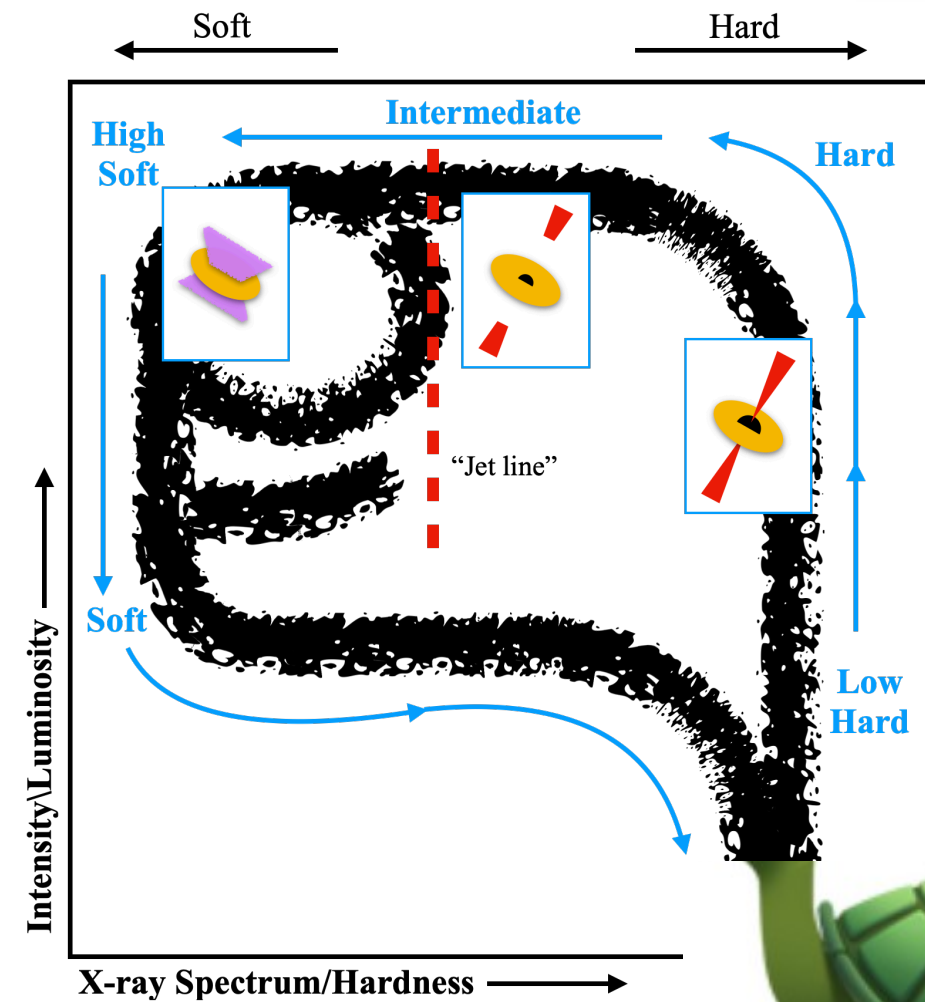
Credit: NASA's Goddard Space
Flight Center/Jeremy Schnittman

Neutron stars

- Complicated structure and basically unknown equation of state
- Hard surface for accretion flow to hit and form hot shocks
- High magnetic fields; pulsations are usually precisely measured



Credit: NASA / JPL-Caltech.



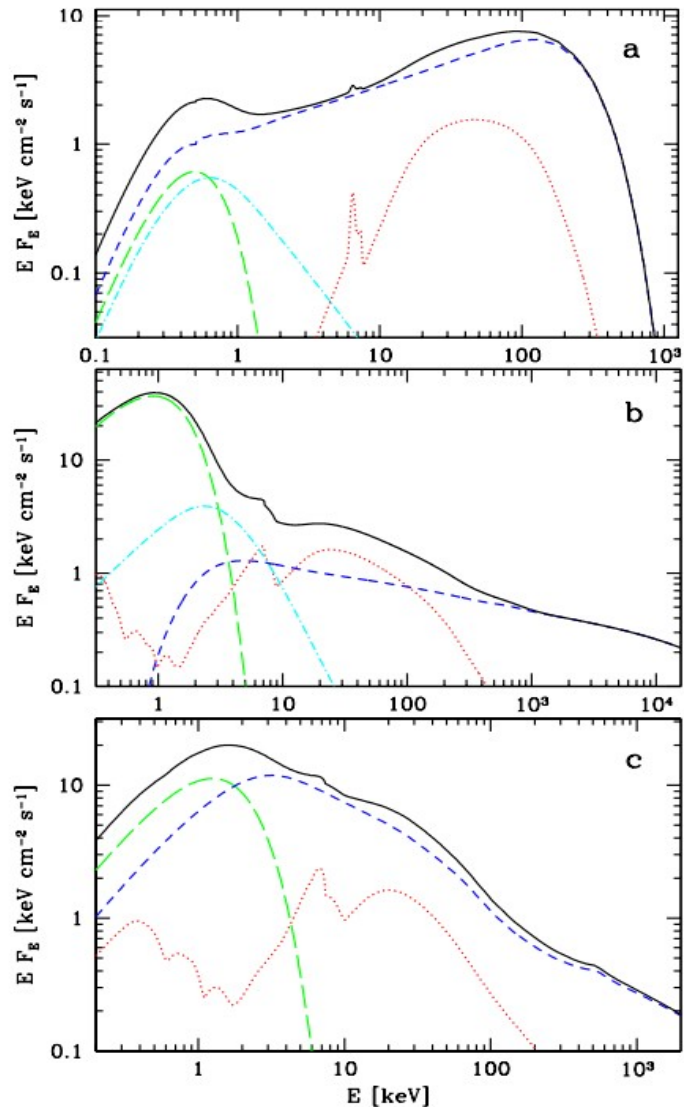
States and transitions

- The XRBs exist in different (spectral) states and transition between them; the balance between different components changes.
- There is a general ‘turtle head’ pattern, but each individual source’s behavior is unique
- XRBs as a class are very multidimensional in parameter space



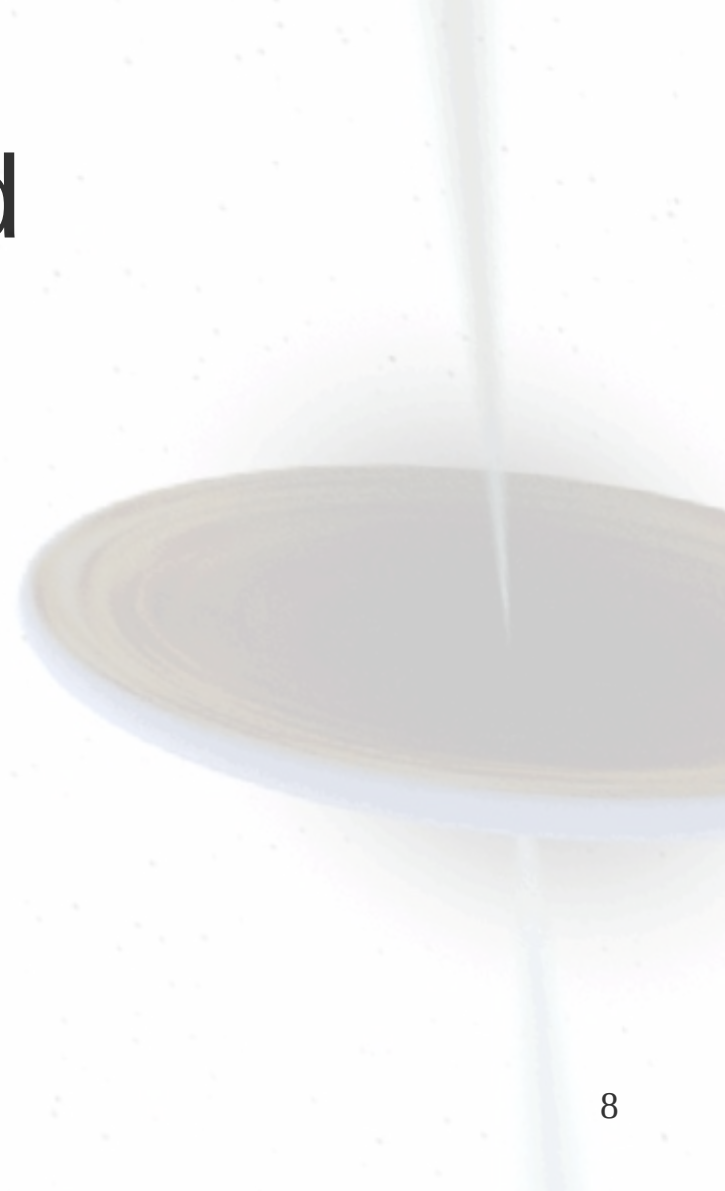
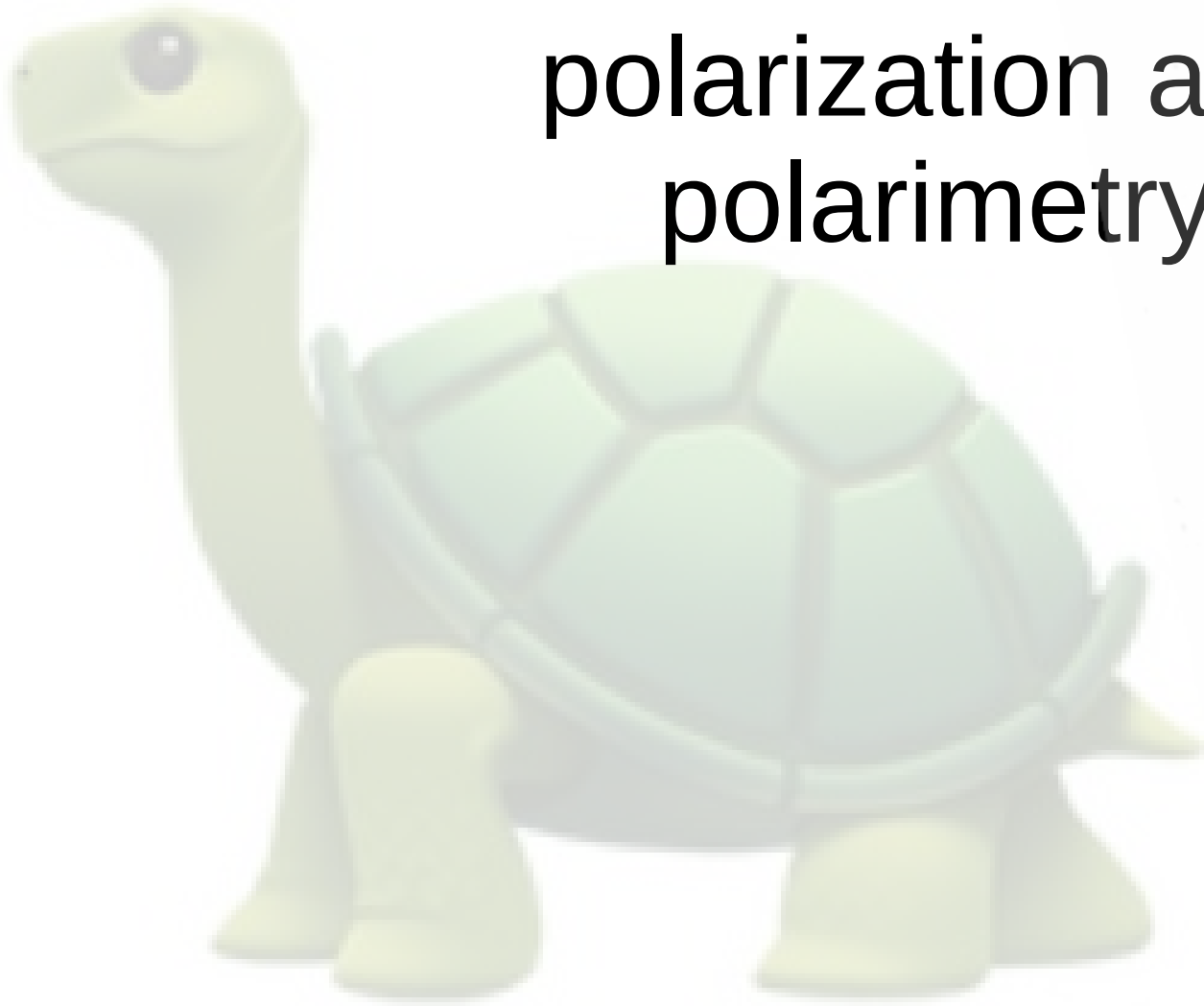
[Moravec et al. 2023]

States and transitions

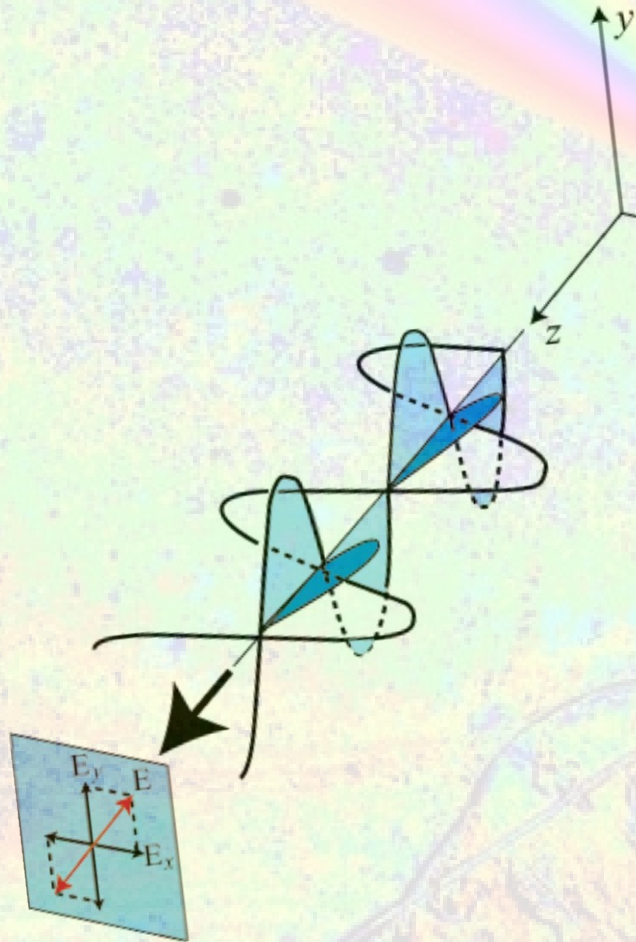


- Supposedly, different components of the system have their place in the spectrum.
- In spectral data, there's no image, no 'depth', little information about geometry.
- Spectra are fundamentally one-dimensional and decomposing and interpreting them is a tricky task.

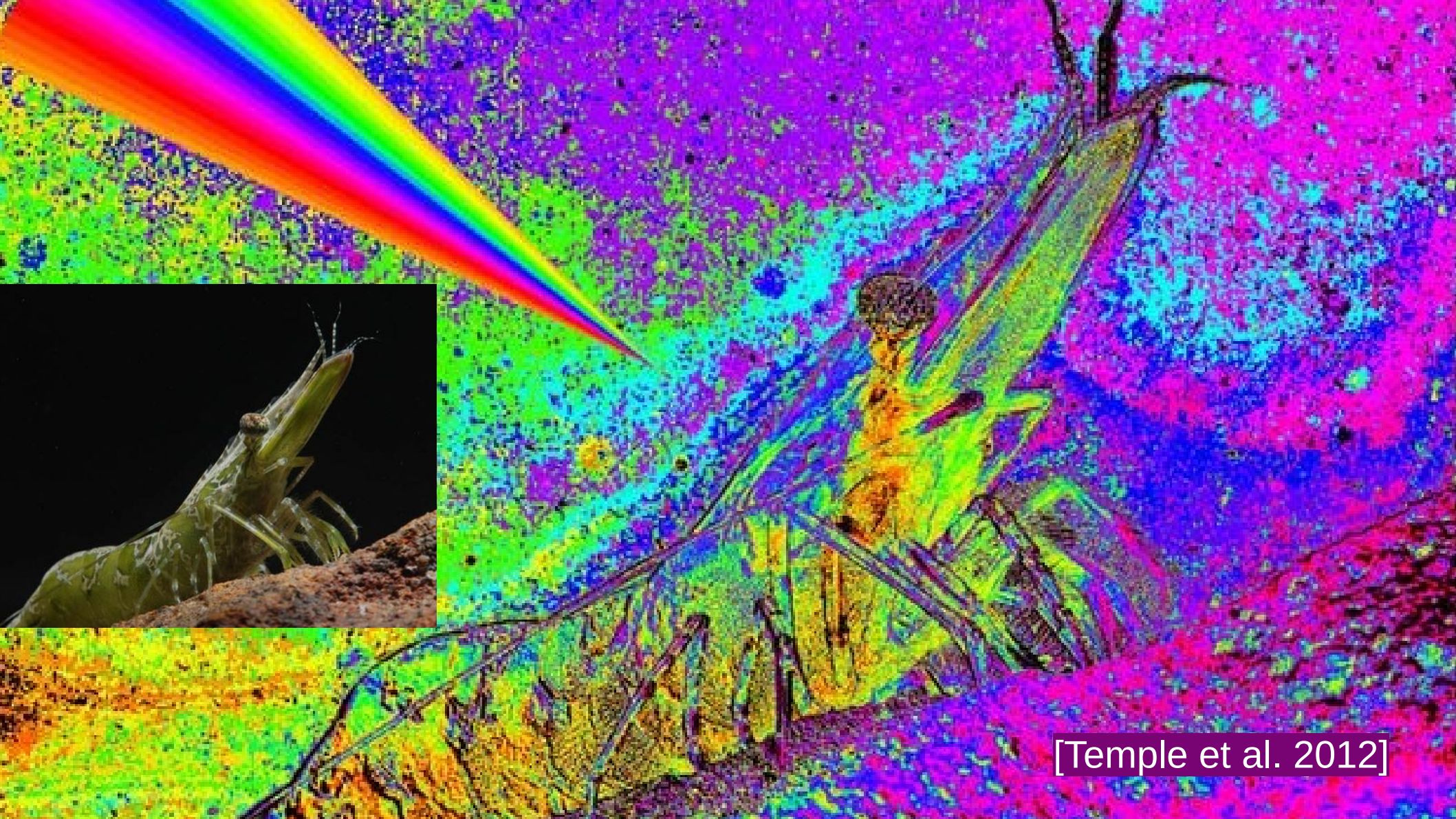
Introduction to polarization and polarimetry



Polarization of light (X-rays)



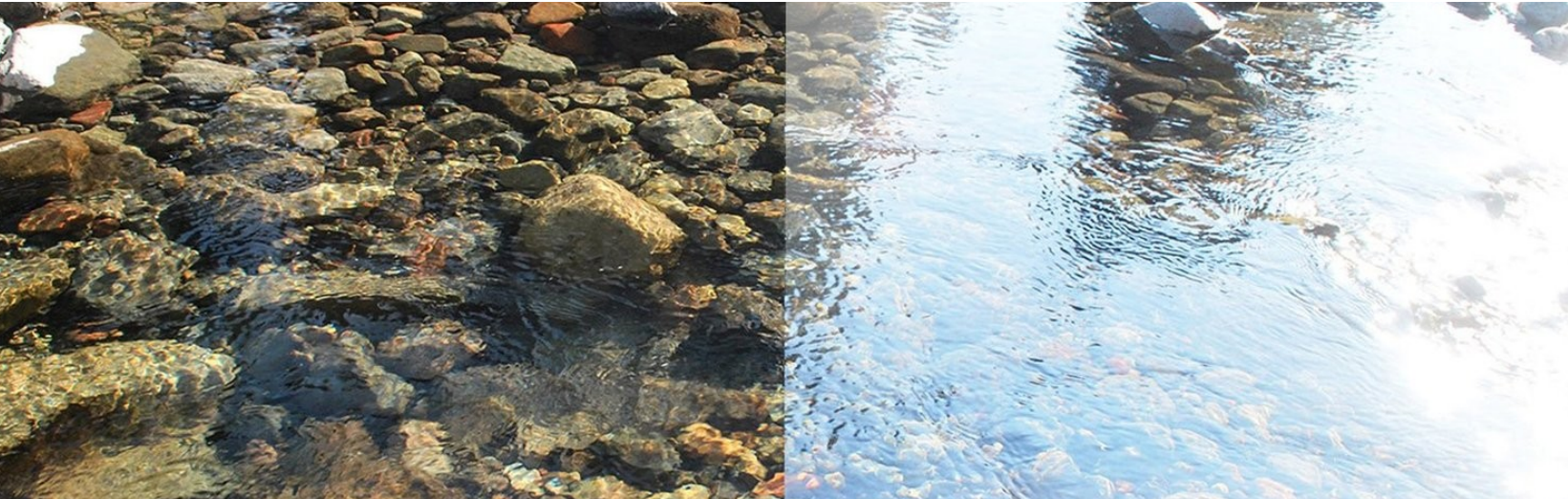
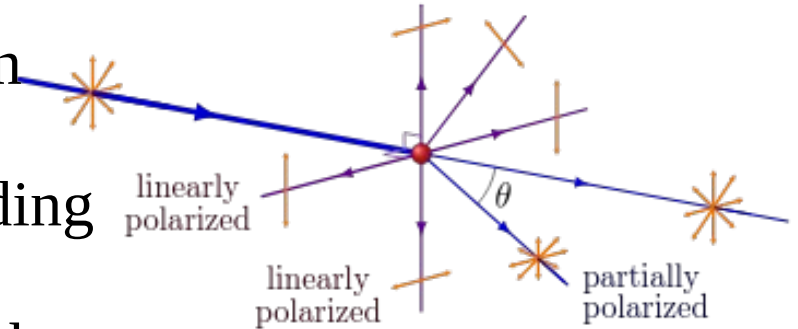
- Not only does an electro-magnetic wave have a wavelength, but also polarization – predominant plane of oscillation
- Polarization is described by two parameters (Stokes parameters, or, alternatively, polarization **angle** and **degree**)
- Essentially, polarization information is two additional spectra and adds two dimensions to the data.



[Temple et al. 2012]

Sources of polarization

- Scattering off particle is very a very common polarization mechanism
- It changes the polarization properties depending on the direction
- Scattering may also change the energy of a photon

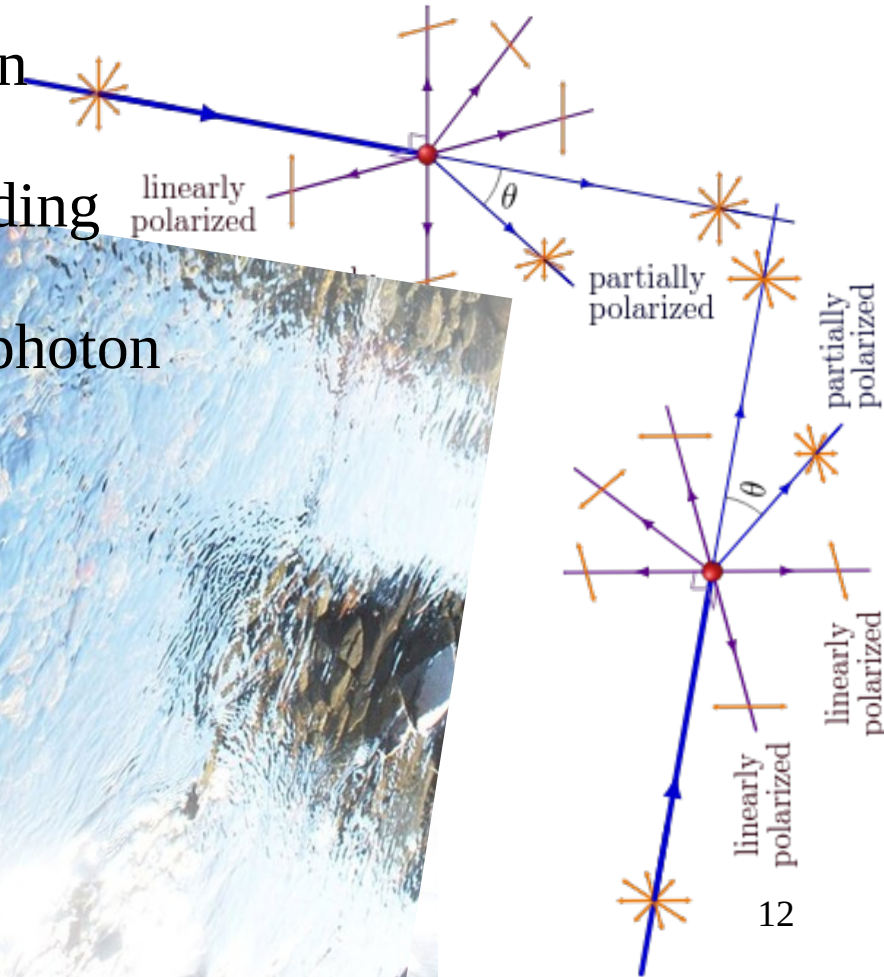
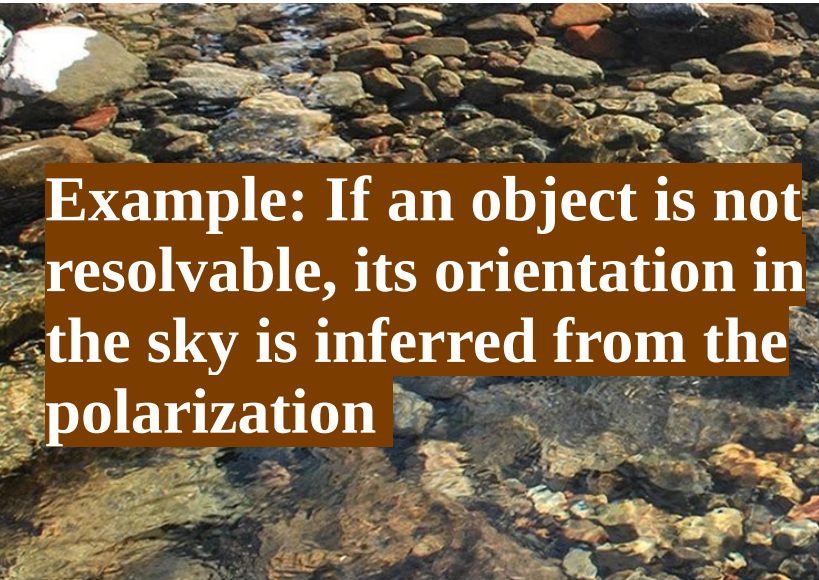


Magnetic fields also alter polarization but let's not focus on them here

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Example: If an object is not resolvable, its orientation in the sky is inferred from the polarization





X-ray polarimetry with IXPE

- Imaging X-ray Polarimetry Explorer
- Launched on December 9th 2021 starting the new era
- A NASA mission in collaboration with the Italian Space Agency (and many researchers from Finland, esp. Quantum 2nd floor)
- > 200 papers with polarization discoveries (including all the brightest XRBs)

Applications and results

Modeling the polarization spectrum

Geometry of the system

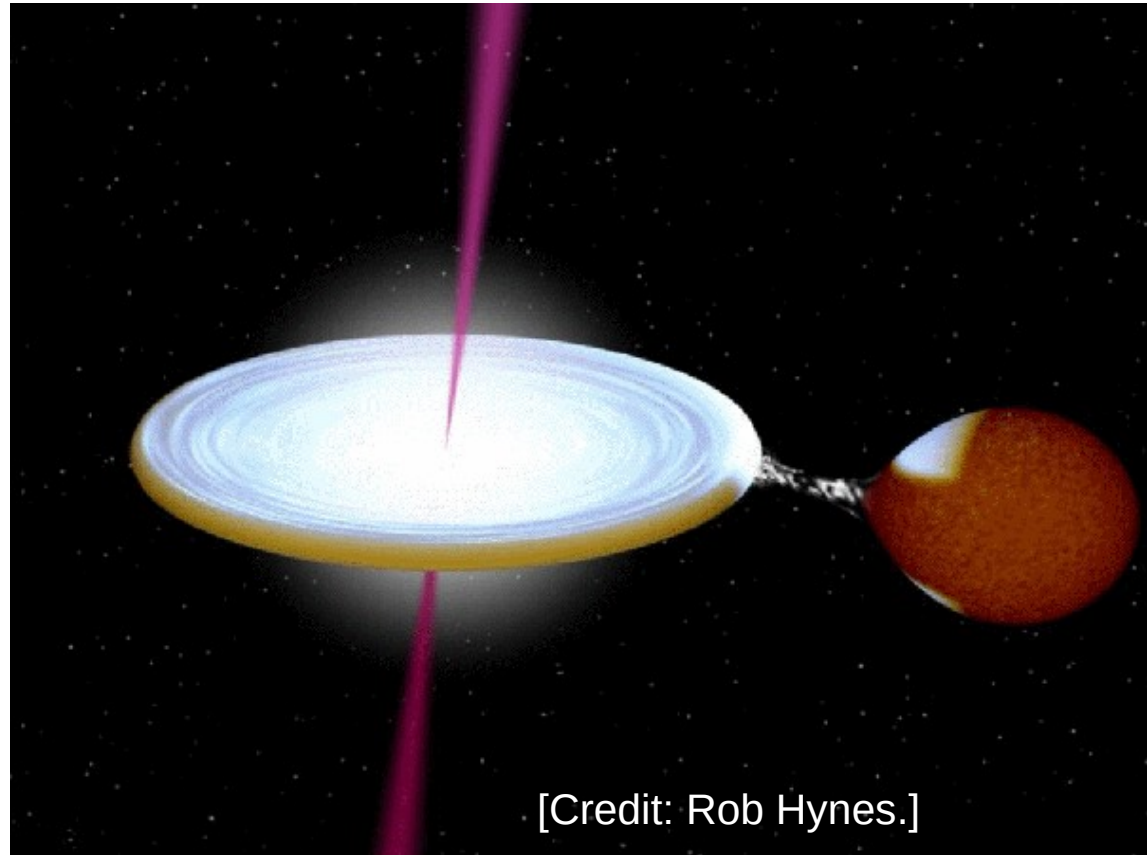
- Spatial relation between different components
- Direction to the observer

Local emission at the source

- PA and PD depend on the perspective
- Different components can

Special and General Relativity

- Relativistic velocities
- Spacetime curvature



[Credit: Rob Hynes.]

To figure out the parameters of a system and the underlying physics, we have to build our best models to compare with observations

Polarization of Millisecond pulsars

Geometry of the system

- Oblate spheroid due to fast rotation

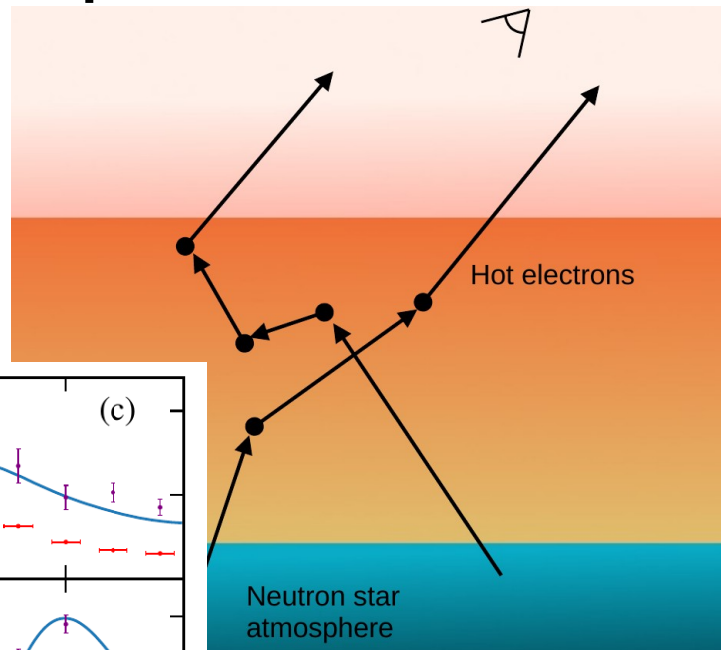
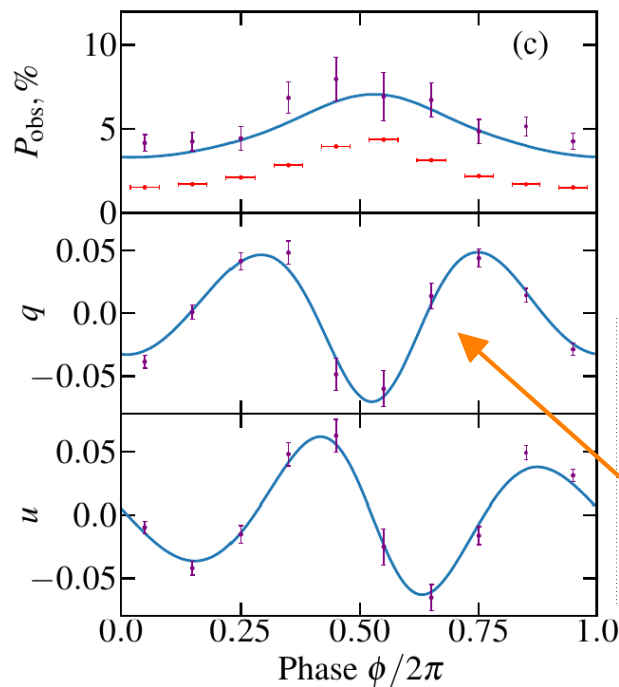
Local emission at the source

- Comptonizing hot electron shock
- Polarization depends on geometrical configuration, resolving degeneracy between the geometrical parameters

Special and General Relativity

- Doppler boosting by rotation ($\sim 10\% c$)
- Schwarzschild metric ($1.4 M_{\text{sol}}$ vs 10-15 km)
- Compared to numerical spacetimes

[Bobrikova et al. 2023]



Modeling the pulse profile and comparing it to observations allows NS parameters inference

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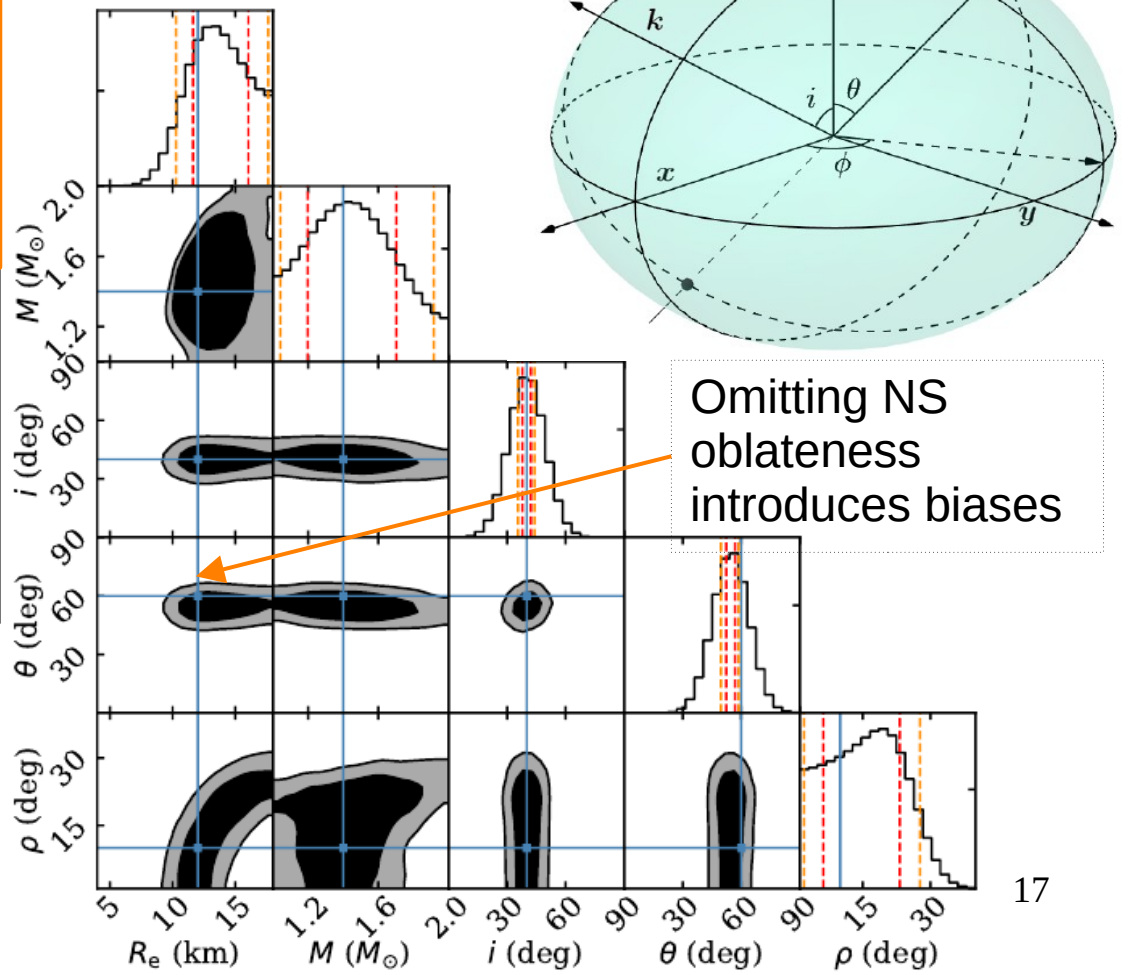
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[Loktev et al. 2020]



Polarization of Black Hole XRBs

Geometry of the system

- Flat equatorial accretion disk
- BH coronae can be of different shapes

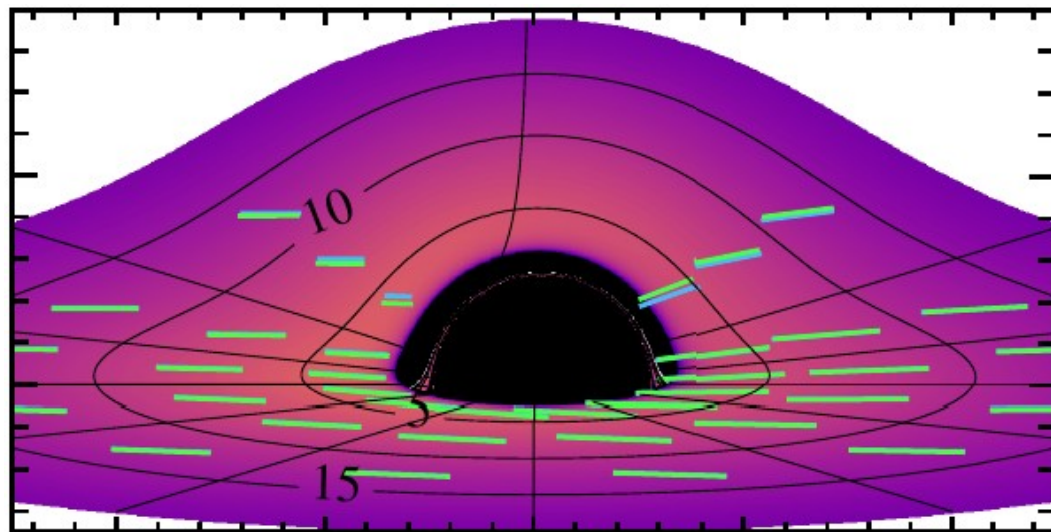
Local emission at the source

- Complex composition of disk atmosphere
- Polarization depends on inclination
- Reprocessed by the BH corona

Special and General Relativity

- Rotating black hole drags spacetime around
- Inner radius of the disk depends on the spin
- Orbital motion is relativistic in the inner part
- Black hole can launch relativistic outflows

- We developed a fast method of computing polarization spectra and images of disks around Kerr BHs
- It is flexible and useful for developing new models for new observations of XRBs, many of which discard previous predictions



[Loktev et al. 2022, 2024]

Polarization of Black Hole XRBs

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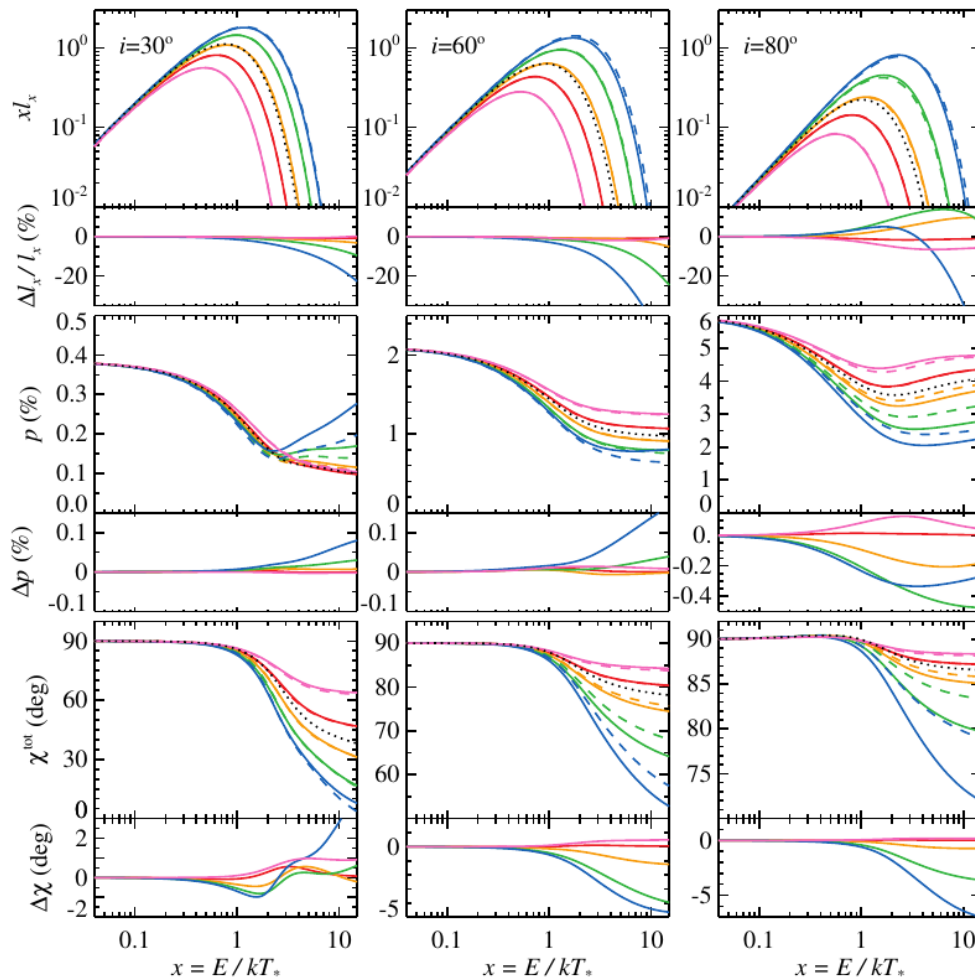
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[Loktev et al. 2022, 2024]

Conclusions

1. XRBs are an insanely diverse class of objects with multiple parameters highlighting significant behavior differences.
2. X-ray polarimetry, recently made possible by IXPE, increases the dimensionality of X-ray data, allowing us to catch up with the complexity of the systems.
3. We have developed several methods and models to infer parameters of XRBs using newly available spectropolarimetric data.

Fig. 5. Spectropolarimetric

A spectro-polarimetric view into accretion plasma physics

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